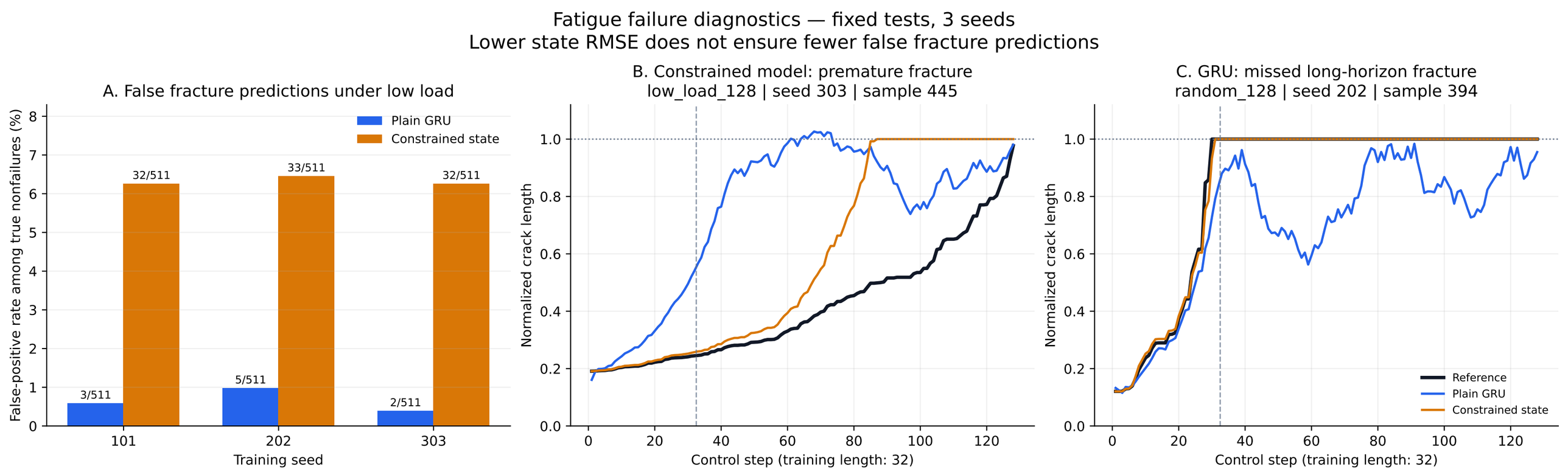
Fatigue failure diagnostics — fixed tests, 3 seeds
Lower state RMSE does not ensure fewer false fracture predictions
A. False fracture predictions under low load
False-positive rate among true nonfailures (%)
Plain GRU
Constrained state
3/511
32/511
5/511
33/511
2/511
32/511
101
202
303
Training seed
B. Constrained model: premature fracture
low_load_128 | seed 303 | sample 445
Normalized crack length
Control step (training length: 32)
C. GRU: missed long-horizon fracture
random_128 | seed 202 | sample 394
Normalized crack length
Control step (training length: 32)
Reference
Plain GRU
Constrained state